

Nicholas Ratner

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EDUCATION

Toronto Metropolitan University

Bachelor of Science in Computer Science, Co-op

2021 – 2026

Toronto, ON

TECHNICAL SKILLS

Languages: Python, Java, C, C#, Bash, HTML/CSS, JavaScript, SQL

Developer Tools: Git, GitHub, Linux, Azure, GitHub Actions, Docker, REST API, VS Code, Visual Studio, MySQL, Oracle, Power BI

Libraries: Pandas, NumPy, Django, PySpark, Qdrant, Paramiko, Ansible

EXPERIENCE

Data Center GPU Validation Engineer

AMD

May 2025 – August 2026

Markham, ON

- Built **Python** packages and **Bash** tools to automate provisioning, validation, and diagnosis of AMD datacenter GPUs.
- Integrated **LLMs** and **vector-database RAG** into a **Python** app to provide automated GPU issue diagnosis and remediation.
- Built **CI/CD** workflows with **GitHub Actions** to automate testing and validation across various **Linux** environments.
- Contributed to a full-stack **Django** dashboard for monitoring GPU test jobs and executing job actions, working on **front-end** development, **back-end** services, and **REST API** design and integration.

Data Engineer

Workplace Safety and Insurance Board (WSIB)

May 2024 – December 2024

Toronto, ON

- Developed **Python** scripts to automate common business tasks, reducing manual effort by **40%** and improving efficiency across various departments.
- Utilized **Python** and **Apache Spark** in **Azure Synapse** notebooks to process and analyze databases with over **10 million** records.
- Leveraged a suite of Azure cloud products, including **Azure Data Lake Storage** and **Azure Blob Storage**, to design and implement data pipelines and storage solutions.
- Created **SQL** queries to efficiently extract and transform data from large-scale databases with over **100 million rows** while improving the efficiency of retrieval by over **50%**.
- Designed **Power BI** dataflows and dashboards, enhancing data accessibility and visualization across 5+ departments, enabling more informed business decisions.

PROJECTS

Concert Database System | *HTML/CSS, JavaScript, PHP, MySQL, Linux/Bash*

- * Worked in a **team environment** to create a simulation concert ticket management system.
- * Utilized **MySQL database** to store and retrieve all necessary data for the system.
- * Created both a **GUI** and a **command-line interface** for users to interact with the system.
- * Collaborated with peers by utilizing **Git** for version control.

Multiplayer Chess Game with AI | *Python*

- * Created an interactive chess game using various Python libraries including **TKinter**, **Pygame**, and **PIL**.
- * Developed a multiplayer mode allowing players to play against one another locally. The UI includes a user-friendly design, music, sound effects, and various quality-of-life features such as allowing users to undo a move and reset the game.
- * Engineered the AI component of the game using the **Minimax** algorithm with **Alpha-Beta pruning**.
- * Using a complex heuristic function the AI can evaluate any board and calculate several moves ahead, finding the optimal one to play.